The Integers and Division, Primes, GCD, LCM

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Outline

1. Division
2. Modular Arithmetic
3. Application of Congruences
4. Primes and Composites
5. GCD and LCM
6. Summary

The Division Algorithm

Theorem (Division Algorithm)

Let a be an integer and d a positive integer. Then there exist unique integers q and r such that

$$a = dq + r, \quad 0 \leq r < d.$$

Terminology of Division

In the equation $a = dq + r$:

- d is the **divisor**
- a is the **dividend**
- q is the **quotient**
- r is the **remainder**

Notation:

$$q = a \text{ div } d, \quad r = a \bmod d$$

Example – Positive and Negative Dividend

Positive Dividend

Problem: Divide 101 by 11.

$$101 = 11 \cdot 9 + 2$$

- Quotient: $101 \text{ div } 11 = 9$
- Remainder: $101 \bmod 11 = 2$

Negative Dividend

Problem: Divide -11 by 3.

$$-11 = 3(-4) + 1$$

- Quotient: $-11 \text{ div } 3 = -4$
- Remainder: $-11 \bmod 3 = 1$

Modular Arithmetic

Definition

Let a and b be integers and m a positive integer. We say that a is **congruent to** b modulo m if m divides $(a - b)$.

$$a \equiv b \pmod{m}$$

If not congruent:

$$a \not\equiv b \pmod{m}$$

Characterization Using Remainders

Theorem 3

Let a and b be integers and m a positive integer.

$$a \equiv b \pmod{m} \iff a \bmod m = b \bmod m$$

Example of Congruence

Example of Congruence

- $17 - 5 = 12$, and $6 \mid 12$

$$\Rightarrow 17 \equiv 5 \pmod{6}$$

- $24 - 14 = 10$, and $6 \nmid 10$

$$\Rightarrow 24 \not\equiv 14 \pmod{6}$$

Theorem 4: Linear Form of Congruence

Let m be a positive integer. Integers a and b are congruent modulo m if and only if there exists an integer k such that

$$a = b + km$$

Operations with Congruence

Theorem 5

Let m be a positive integer. If

$$a \equiv b \pmod{m}, \quad c \equiv d \pmod{m}$$

Then:

$$a + c \equiv b + d \pmod{m}$$

$$ac \equiv bd \pmod{m}$$

Example of Modular Operations

Given:

$$7 \equiv 2 \pmod{5}, \quad 11 \equiv 1 \pmod{5}$$

Then:

$$7 + 11 = 18 \equiv 3 \pmod{5}$$

$$7 \cdot 11 = 77 \equiv 2 \pmod{5}$$

Useful Modular Identities

Corollary

Let m be a positive integer and a, b integers.

$$(a + b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$$

$$(ab) \bmod m = ((a \bmod m)(b \bmod m)) \bmod m$$

Example Using Identities

$$(8 + 14) \bmod 5 = 22 \bmod 5 = 2$$

$$((8 \bmod 5) + (14 \bmod 5)) \bmod 5 = (3 + 4) \bmod 5 = 2$$

$$(8 \cdot 14) \bmod 5 = 112 \bmod 5 = 2$$

Pseudorandom Numbers

Pseudorandom Numbers

A sequence of numbers generated by a deterministic mathematical formula but exhibiting random-like behavior is called a **pseudorandom number sequence**.

Definition

The most commonly used method for generating pseudorandom numbers is the **linear congruential method**.
We choose four integers:

- modulus m
- multiplier a
- increment c
- seed x_0

subject to:

$$2 \leq a < m, \quad 0 \leq c < m, \quad 0 \leq x_0 < m$$

Linear Congruential Method

Recurrence Relation

A sequence $\{x_n\}$ is generated by:

$$x_{n+1} = (ax_n + c) \bmod m$$

where $0 \leq x_n < m$ for all n.

Example: Pseudorandom Number Generation

Given Parameters

Let:

$$m = 9, \quad a = 7, \quad c = 4, \quad x_0 = 3$$

Recurrence Formula

$$x_{n+1} = (7x_n + 4) \bmod 9$$

Step-by-Step Computation

$$x_0 = 3$$

$$x_1 = (7 \cdot 3 + 4) \bmod 9 = 25 \bmod 9 = 7$$

$$x_2 = (7 \cdot 7 + 4) \bmod 9 = 53 \bmod 9 = 8$$

$$x_3 = (7 \cdot 8 + 4) \bmod 9 = 60 \bmod 9 = 6$$

Generating the Sequence

Continued Computation

$$x_4 = (7 \cdot 6 + 4) \bmod 9 = 46 \bmod 9 = 1$$

$$x_5 = (7 \cdot 1 + 4) \bmod 9 = 11 \bmod 9 = 2$$

$$x_6 = (7 \cdot 2 + 4) \bmod 9 = 18 \bmod 9 = 0$$

$$x_7 = (7 \cdot 0 + 4) \bmod 9 = 4$$

$$x_8 = (7 \cdot 4 + 4) \bmod 9 = 32 \bmod 9 = 5$$

$$x_9 = (7 \cdot 5 + 4) \bmod 9 = 39 \bmod 9 = 3$$

Observations

Key Observations

- The sequence is completely determined by the initial seed x_0
- The values repeat after a finite number of steps
- The maximum possible period is m
- The numbers are not truly random, but appear random

Prime and Composite Numbers

Definition

A positive integer $p > 1$ is called **prime** if its only positive divisors are 1 and p.

A positive integer greater than 1 that is not prime is called **composite**.

Examples

- 7 is prime since its only positive divisors are 1 and 7
- 9 is composite since it is divisible by 3

Fundamental Theorem of Arithmetic

Theorem 1

Every positive integer greater than 1 can be written uniquely as a prime or as a product of two or more primes, where the prime factors are written in non-decreasing order.

Examples

$$100 = 2^2 \cdot 5^2$$

$$641 = 641$$

$$999 = 3^3 \cdot 37$$

$$1024 = 2^{10}$$

Testing Primality

Theorem 2

If n is a composite integer, then n has at least one prime divisor less than or equal to $\sqrt{n}$.

Example

Show that 101 is prime.:

Primes $\leq \sqrt{101} = 10$ are:

$$2, 3, 5, 7$$

Since 101 is not divisible by any of these, 101 is prime.

Example

Find the prime factorization of 7007.

$$7007 = 7 \cdot 1001 = 7 \cdot 7 \cdot 143 = 7^2 \cdot 11 \cdot 13$$

Greatest Common Divisor (GCD)

Definition

Let a and b be integers, not both zero. The largest integer that divides both a and b is called the **greatest common divisor**.

$$\gcd(a, b)$$

Example

$$\gcd(24, 36) = 12$$

$$\gcd(17, 22) = 1$$

Relatively Prime Integers

Definition: Relatively Prime Integers

Integers a and b are **relatively prime** if

$$\gcd(a, b) = 1$$

Example

17 and 22 are relatively prime.

Definition: Pairwise Relatively Prime Integers

Integers $a_1, a_2, \ldots, a_n$ are **pairwise relatively prime** if

$$\gcd(a_i, a_j) = 1 \quad \text{for all } i < j$$

Example

- 10, 17, and 21 are pairwise relatively prime
- 10, 19, and 24 are not, since $\gcd(10, 24) = 2$

Computing GCD Using Prime Factorization

Method

Suppose:

$$a = p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n}$$

$$b = p_1^{b_1} p_2^{b_2} \cdots p_n^{b_n}$$

where each p_i is prime and exponents are nonnegative integers. Then:

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \cdots p_n^{\min(a_n, b_n)}$$

Example

Find $\gcd(120, 500)$.

$$120 = 2^3 \cdot 3^1 \cdot 5^1$$

$$500 = 2^2 \cdot 3^0 \cdot 5^3$$

$$\gcd(120, 500) = 2^{\min(3,2)} 3^{\min(1,0)} 5^{\min(1,3)} = 2^2 \cdot 3^0 \cdot 5^1 = 20$$

Least Common Multiple (LCM)

Definition

The **least common multiple** of two nonzero integers a and b, denoted by $\text{lcm}(a, b)$, is the smallest positive integer divisible by both a and b.

Prime Factorization Method

Suppose that the prime factorizations of the integers a and b, neither equal to zero, are:

$$a = p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n}$$
$$b = p_1^{b_1} p_2^{b_2} \cdots p_n^{b_n}$$

where each exponent is a nonnegative integer, and where all primes occurring in the prime factorization of either a or b are included in both factorizations (with zero exponents if necessary). Then

$$\text{lcm}(a, b) = p_1^{\max(a_1, b_1)} p_2^{\max(a_2, b_2)} \cdots p_n^{\max(a_n, b_n)}$$

Computing LCM Using Prime Factorization

Example

Find $\text{lcm}(95256, 432)$.

$$95256 = 2^3 \cdot 3^5 \cdot 7^2$$

$$432 = 2^4 \cdot 3^3 \cdot 7^0$$

$$\text{lcm}(95256, 432) = 2^{\max(3,4)} \cdot 3^{\max(5,3)} \cdot 7^{\max(2,0)}$$

$$= 2^4 \cdot 3^5 \cdot 7^2 = 190512$$

Summary

- Division Algorithm explains quotient and remainder
- Modular arithmetic simplifies integer computations
- Primes are the building blocks of integers
- GCD and LCM describe divisibility relationships

End of Lecture

Thank You